

L&T EduTech

Applied Industrial Internet of Things

Project Report

**IoT-enabled Gas Leak and Fire Detection
System using Raspberry Pi Pico W**

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Abstract

This project presents an IoT-enabled Gas Leak and Fire Detection System using Raspberry Pi Pico W. The system continuously monitors gas leakage and fire using an MQ-2 Gas Sensor and a Flame Sensor. When hazardous conditions are detected, a buzzer and LED are activated to provide an instant alert. At the same time, sensor data is transmitted to an IoT dashboard for real-time monitoring. This solution is affordable, reliable, and suitable for homes, laboratories, industries, and commercial buildings.

Problem Statement

Gas leakage and fire accidents can cause serious damage to life and property. Traditional monitoring systems often fail to provide real-time alerts and remote monitoring. Therefore, an IoT-enabled system is required to detect gas leakage and fire at an early stage, generate instant alerts, and send live data to an IoT dashboard for continuous monitoring and quick emergency response.

Objectives

- To detect gas leakage using the MQ-2 Gas Sensor.
 - To detect fire using the Flame Sensor.
 - To provide instant alerts using a buzzer and LED.
 - To send real-time sensor data to an IoT dashboard.
 - To improve safety in homes, industries, and laboratories.
-

Scope of the Solution

The proposed system continuously monitors gas leakage and fire conditions in real time. If the gas concentration exceeds the threshold or fire is detected, the system immediately activates an alarm and updates the IoT dashboard. This project can be used in homes, industries, laboratories, LPG storage areas, restaurants, and warehouses. Future improvements include SMS alerts, email notifications, mobile application integration, automatic exhaust fan control, and fire sprinkler systems.

Required Components

Hardware

- Raspberry Pi Pico W
- MQ-2 Gas Sensor
- Flame Sensor
- Buzzer
- LED
- Breadboard
- Jumper Wires
- USB Cable
- Wi-Fi Connection

Software

- MicroPython
- GitHub

Programming Language

MicroPython

Circuit Diagram

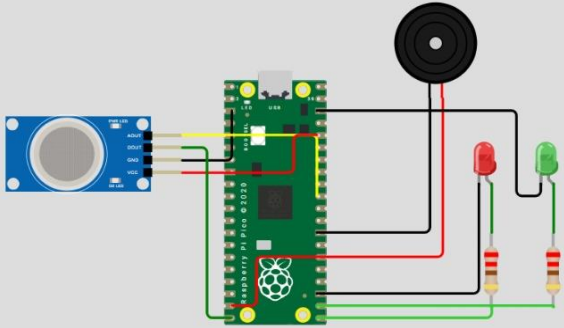
WOKWI

SAVE SHARE Arun thakur

sketch.ino diagram.json Library Manager

```
1 const int gasSensor = A0; // GP26
2 const int redLED = 16; // GP16
3 const int greenLED = 17; // GP17
4
5 void setup() {
6   Serial.begin(115200);
7   pinMode(redLED, OUTPUT);
8   pinMode(greenLED, OUTPUT);
9 }
10
11 void loop() {
12   int gasValue = analogRead(gasSensor);
13
14   Serial.print("Gas Value: ");
15   Serial.println(gasValue);
16
17   if (gasValue > 500) {
18     digitalWrite(redLED, HIGH);
19     digitalWrite(greenLED, LOW);
20     Serial.println("Gas Leak Detected!");
21   } else {
22     digitalWrite(redLED, LOW);
23     digitalWrite(greenLED, HIGH);
24     Serial.println("Gas Level Normal");
25   }
26
27   delay(500);
28 }
```

Simulation



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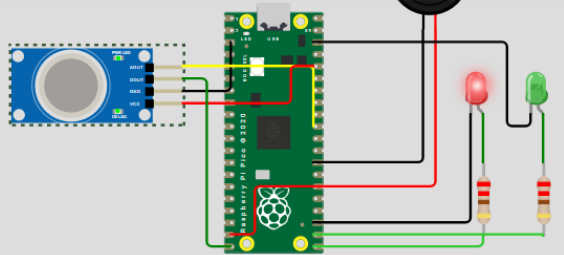
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Simulation

Gas Sensor

GAS (PPM) 219

219 ppm



Code Logic

1. Initialize Raspberry Pi Pico W.
2. Read data from the MQ-2 Gas Sensor.
3. Read data from the Flame Sensor.
4. Compare sensor values with predefined thresholds.
5. If gas leakage or fire is detected:
 - Turn ON the LED.
 - Activate the buzzer.
 - Send data to the IoT dashboard.
6. Otherwise:
 - Keep the system in Safe Mode.
7. Repeat the monitoring process continuously.

Code for the solution

```
const int gasSensor = A0;      // GP26

const int redLED = 16;         // GP16
const int greenLED = 17;      // GP17

void setup() {
    Serial.begin(115200);
    pinMode(redLED, OUTPUT);
    pinMode(greenLED, OUTPUT);
}

void loop() {
    int gasValue = analogRead(gasSensor);

    Serial.print("Gas Value: ");
    Serial.println(gasValue);

    if (gasValue > 500) {
        digitalWrite(redLED, HIGH);
        digitalWrite(greenLED, LOW);
        Serial.println("Gas Leak Detected!");
    } else {
        digitalWrite(redLED, LOW);
        digitalWrite(greenLED, HIGH);
    }
}
```

```
    Serial.println("Gas Level Normal");  
}  
  
delay(500);  
}
```

The solution is developed using **MicroPython** in the **Thonny IDE**. The program continuously reads data from the MQ-2 Gas Sensor and Flame Sensor. If the gas level exceeds the predefined threshold or fire is detected, the Raspberry Pi Pico W activates the buzzer and LED to alert the user. The program runs continuously to provide real-time monitoring and ensure safety.

IoT Dashboard Analytics

The current version of the project focuses on gas leak and fire detection using Raspberry Pi Pico W. The hardware successfully detects hazardous conditions and generates alerts through the LED and buzzer.

The IoT dashboard integration is planned as a future enhancement. In the next version, the system will transmit real-time sensor data to an IoT cloud platform such as Blynk IoT or ThingSpeak using the Wi-Fi capability of Raspberry Pi Pico W.

Proposed Dashboard Parameters

- Gas Sensor Value
- Flame Sensor Status
- Alert Status (ON/OFF)
- System Status (Safe/Danger)
- Date and Time of Events

Current Status: *IoT dashboard integration has not been implemented in this project version.*

Result

The developed system successfully detects gas leakage and fire hazards using the MQ-2 Gas Sensor and Flame Sensor.

Whenever gas leakage or fire is detected, the Raspberry Pi Pico W activates the buzzer and LED to provide an immediate alert.

The system operates reliably and demonstrates an effective solution for improving safety in homes, laboratories, and industries.

